

Pio Canas Toimil

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PROFESSIONAL SUMMARY

Computer Science with High Performance Graphics and Games Engineering graduate from the University of Leeds with an Upper Second-Class Honours (2:1), experienced in developing graphics and engine systems for a custom C++ game engine. Proficient in Vulkan, OpenGL, and real-time rendering techniques, with experience developing graphics and engine systems. Seeking a graduate role in graphics, rendering, or game engine programming.

PROJECTS

KillSwitch Engine – Master’s Group Project, Feb 2026 – May 2026

- Developed core game engine systems in C++, including asset pipeline, entity system, and skeletal animation framework as part of a five-person team.
- Designed and implemented tooling to convert glTF and GLB assets into runtime resources used by the engine.
- Collaborated on a third-person horror demo game called *Hell Mist* to showcase the engine at the 2026 Game Republic Student Showcase, where it won the Red Kite Games Award for Game Technology.

Vulkan Renderer – University Project, 2025–2026

- Implemented a real-time renderer using Vulkan and GLSL shaders, developing physically based rendering techniques, texture filtering, mipmapping and post-processing effects.
- Built debug visualisations for mipmap levels, fragment depth, partial derivatives, overdraw and overshading, mesh density, and a toggle for a mosaic/pixelated filter.

Raytracer – University Project, 2025–2026

- Developed a physically based raytracer, implementing ray-triangle intersection, shading and recursive rays.
- Built support for reflections, refraction, Fresnel effect, transparency and soft shadows using area-light sampling.
- Implemented Monte Carlo path tracing with next-event estimation, anti-aliasing and caustics to simulate realistic global illumination.

EDUCATION

MEng Computer Science with High-Performance Graphics and Games Engineering, University of Leeds | 2022–2026

- **Grade:** Upper Second-Class Honours (2:1)

UTC Leeds | 2018 – 2022

- **A-Levels:** Computer Science (A)
- **OCR Cambridge Technical Level 3 Engineering:** Distinction* Distinction*
- **GCSEs:** 9 GCSEs, including Computer Science (8), Mathematics (7), English Language (8).
- **OCR Level 1/2 Cambridge National Certificate** in Engineering Manufacture (Distinction) and Design (Merit)

WORK EXPERIENCE

Event Hospitality Assistant, Xenia Recruitment, Leeds | 2023–2025

- Worked at large sporting and entertainment events, collaborating with teams in fast-paced environments.
- Adapted quickly to the changing demands whilst balancing work with a full-time degree.

INTERESTS AND SKILLS

- **Basketball:** Member of the University of Leeds Basketball Team (2022–2026)
- **Game Development:** Interested in building immersive experiences and exploring real-time rendering technologies and virtual reality.
- **Technical Skills:** C++, C, C#, Python, Java, SQL, Bash | Vulkan, OpenGL, GLSL, Unity, OpenXR | Git, CMake, Blender